SECTION 08 · TEXT CLUSTERING

Clustering Text with Embeddings and a Local LLM

Find what customers are asking about with no labels: embed, reduce, cluster, then let a local LLM name each group.

sentence-transformers | UMAP | HDBSCAN | Ollama | BERTopic | Banking77

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Unsupervised Learning on Text

Support tickets

thousands a day,
no one has time to tag them

Product reviews

what are people
actually complaining about?

Search logs

what are users
looking for and not finding?

Survey answers

free text that never
fits the tick boxes

Most text arrives with no labels at all.

Clustering finds the themes first; people name and act on them afterwards.

CUSTOMER MESSAGES WITH INTENTS HELD BACK

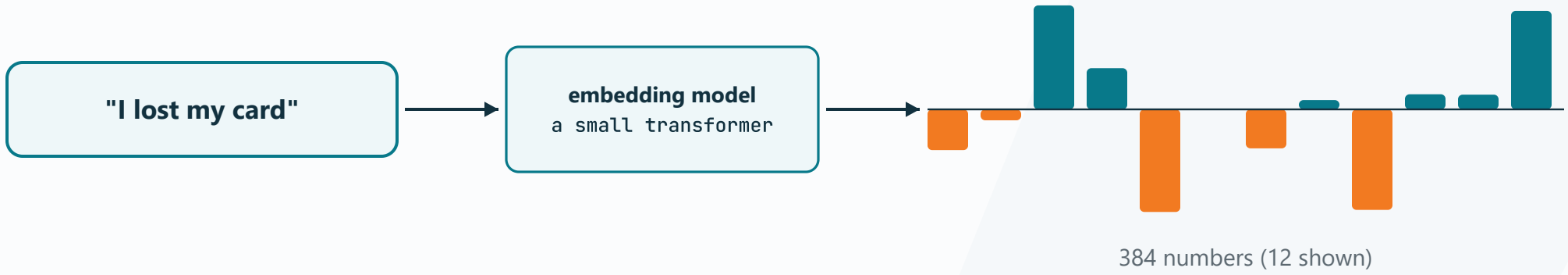
The Banking77 Dataset

message	intent: held back
I am still waiting for my card	`card_arrival`
Why was I charged a fee at the ATM?	`cash_withdrawal_charge`
My PIN is blocked	`pin_blocked`
How do I close my account?	`terminate_account`
The exchange rate looks wrong	`exchange_rate`

77 intents
about 10,000 training messages

The intents play the role of the truth we never had
the clustering never sees them; they only grade it at the end

What an Embedding Is

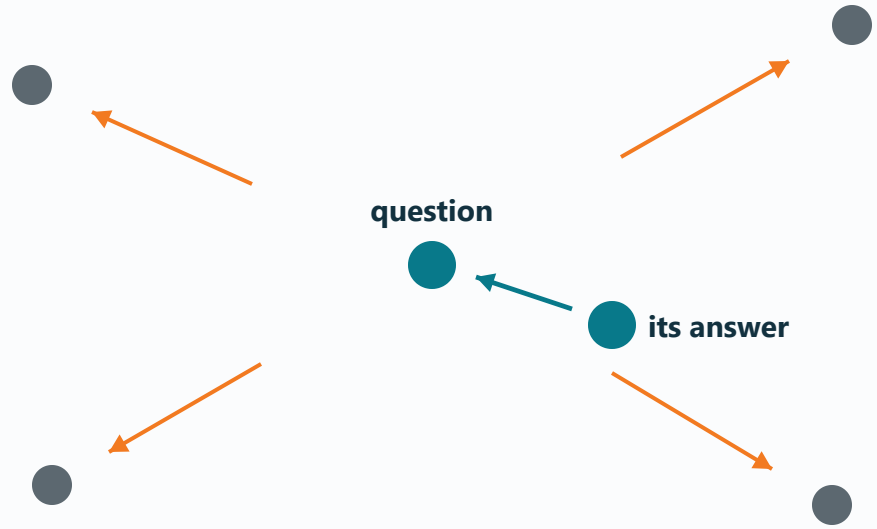


384 numbers (12 shown)

One fixed-length vector per message, whatever its length

messages that mean the same thing land close together, even with no words in common

How Embedding Models Learn



pull pairs together, push everything else away

Contrastive learning

Trained on pairs that belong together: a question and its answer, a title and its article, two wordings of the same request

The pairs come from the web, not from people labelling meaning by hand

Self-supervised, like SASRec
the data supplies its own targets

Cosine Similarity



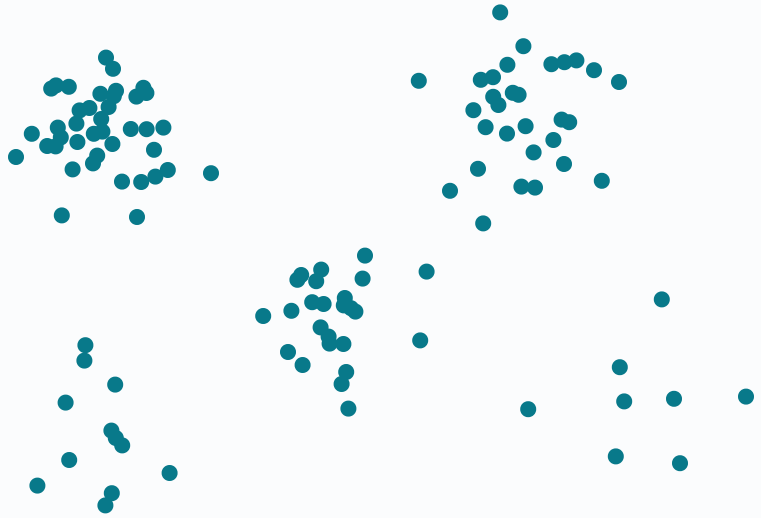
Cosine similarity

the cosine of the angle between two vectors

1: same direction, same meaning
near 0: unrelated

Normalise every vector to length 1
then cosine similarity is a plain dot product, and very fast

k-Means Needs the Number of Intents



how many intents are in here?

k-means must be told k

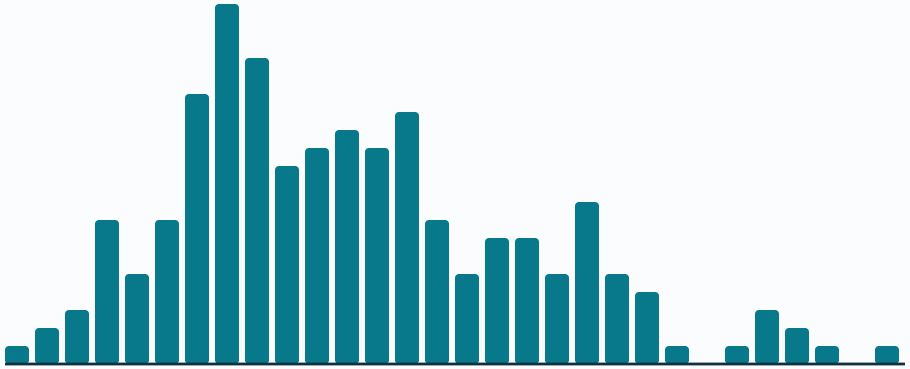
With Banking77 we cheat: the answer is 77
On real tickets nobody knows the number

It also puts every message in a group,
even the ones that fit nowhere

A fair baseline, not a real setting
it gets the true k, which no real project has

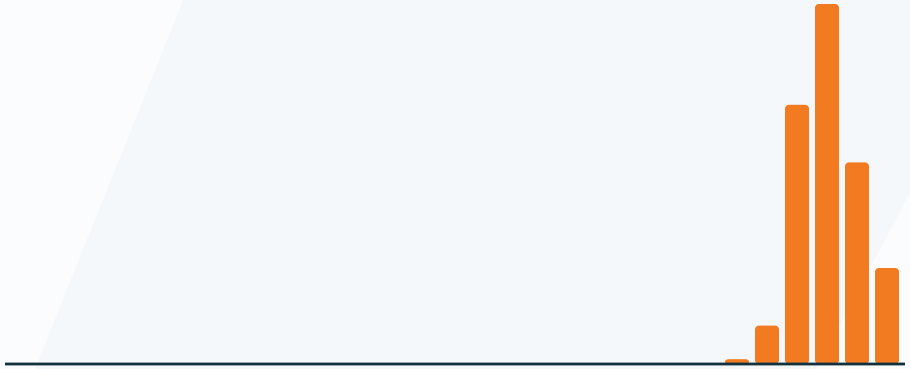
Why Reduce Before Clustering

2 dimensions



distance between random pairs

384 dimensions



distance between random pairs

In many dimensions every distance looks about the same

density has nothing to find, so HDBSCAN calls most points noise

UMAP to a handful of dimensions keeps the neighbours and restores contrast

Embed, Reduce, Cluster, Name

Embed
`sentence-transformers`
meaning as vectors

→

Reduce
`UMAP`
from the dimensionality reduction section

→

Cluster
`HDBSCAN`
from the clustering section

→

Name
`a local LLM`
reads samples, writes a label

Every step is unsupervised or self-supervised
the same pipeline drives topic discovery in most text analytics tools today

Noise Is an Answer



grey: messages between groups

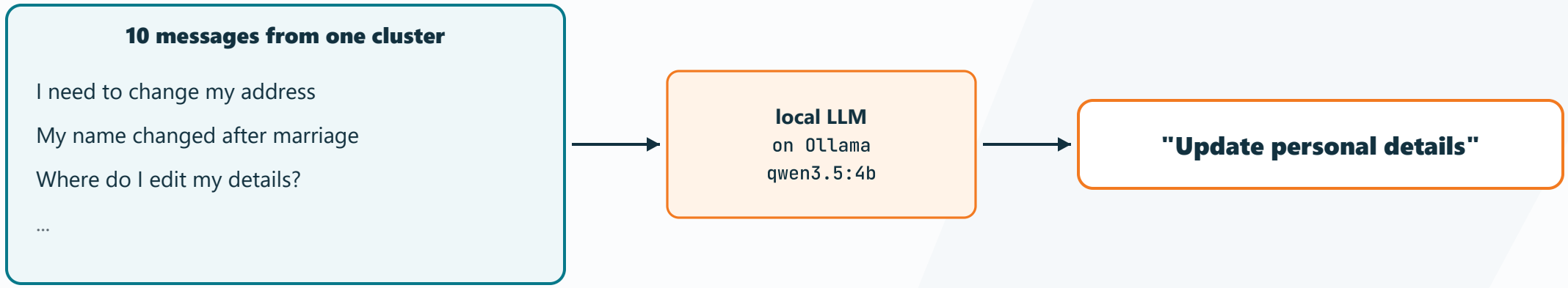
HDBSCAN can say "none of these"

Vague or mixed messages sit between groups, like "is my card lost or late?"

Forcing them into a group would blur that group's meaning and its name

Noise is a queue, not a failure
a person reviews it, or it waits for the next run

Naming Clusters With a Local LLM



Local: customer messages never leave the machine

the prompt asks for a short name only; think=False skips the reasoning text

a person still checks the names: the model only sees ten examples

How BERTopic Names Topics



BERTopic's names: class-based TF-IDF

Treat each topic as one long document
Score words common in this topic but
rare across the others

Fast and free, but only keywords
an LLM turns the same samples into
a name a person would write

Grading Against Held-Back Intents

k-means	told the true number of intents every message placed
UMAP + HDBSCAN	finds the number itself some messages left as noise
BERTopic	the same, packaged plus keyword names

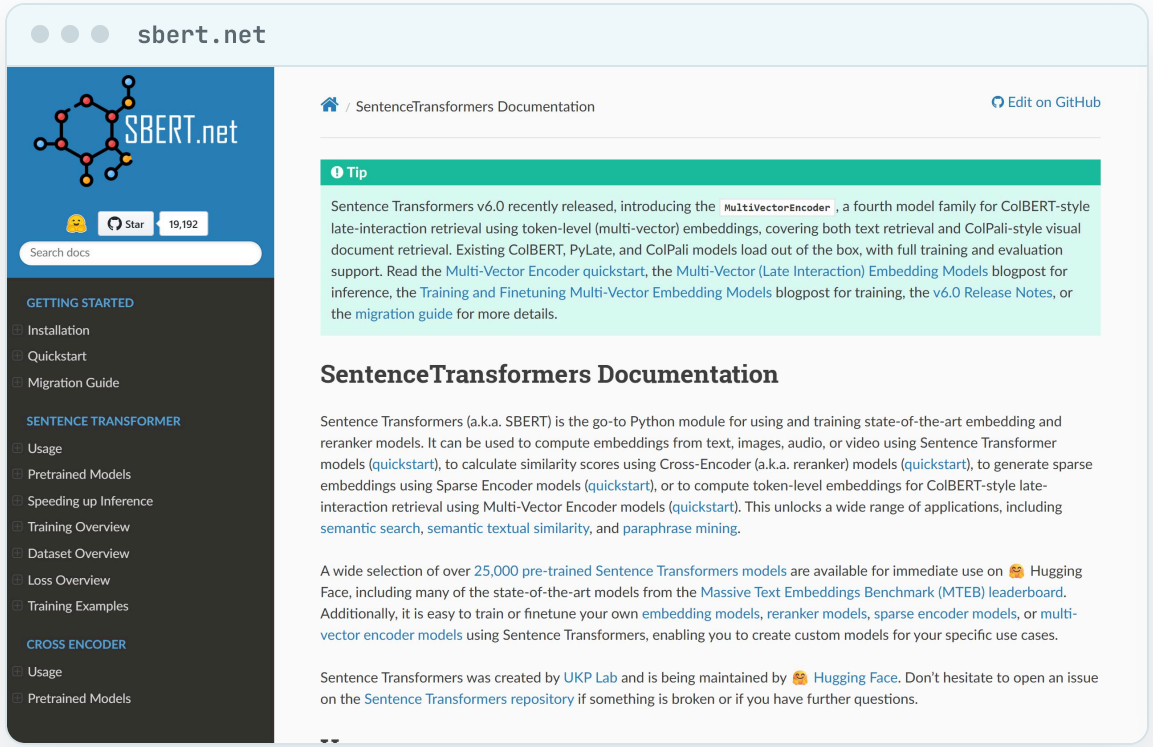
Adjusted Rand index against the intents

Noise counts as one big wrong cluster, so density methods score lower

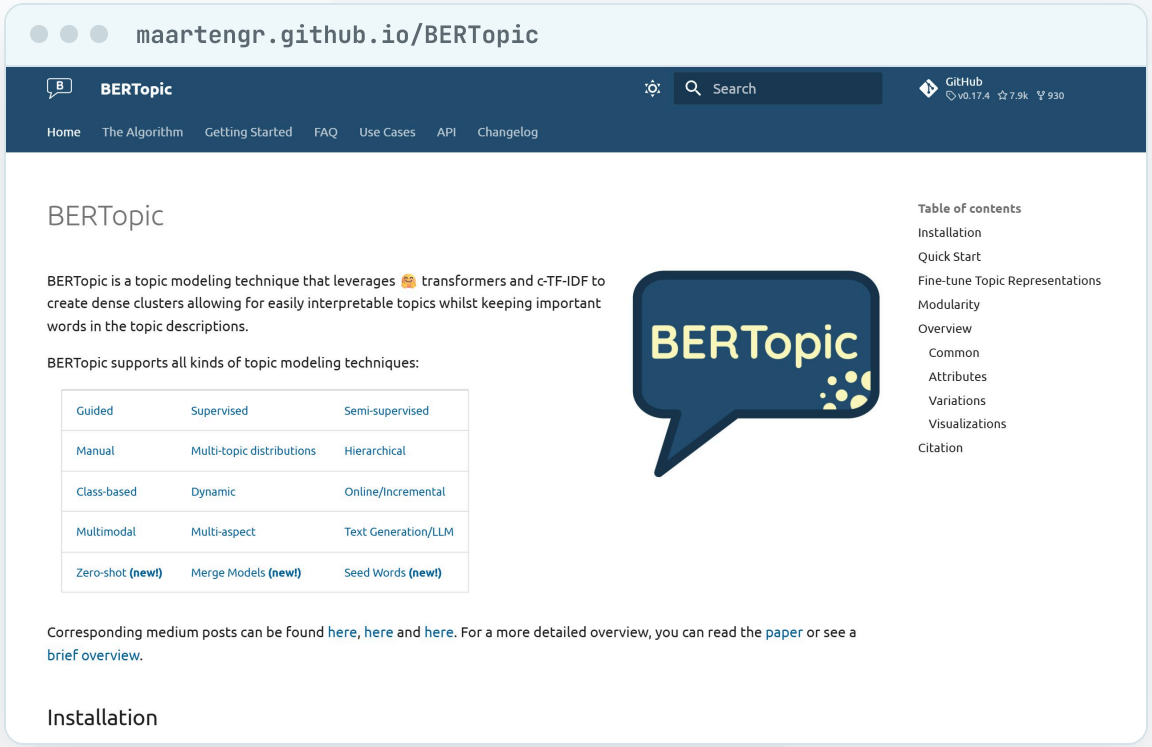
Some intents are near-duplicates, so a merged group can be a fair answer

Read clusters, not just the score
the score cannot tell a good merge from a bad one

Sentence Transformers and BERTopic Docs



Sentence Transformers: embedding models and how to use them



BERTopic: the embed, reduce, cluster pipeline as one model

Both run on a laptop; a GPU only speeds up the embedding step.

From Clusters to a Labelled Dataset

Cluster
unlabelled messages

→

Review
a person names, merges, splits groups

→

Label
cluster names become training labels

→

Classify
a supervised model routes new messages

The semi-supervised idea again
unsupervised learning creates the labels that a cheaper supervised model then uses

Choosing a Text Clustering Setup

STEP	START WITH	CHANGE IT WHEN
Embed	a small sentence-transformers model	the text is long, specialised or not English
Reduce	UMAP, 5 dimensions, cosine	results swing between runs: fix `random_state`
Cluster	HDBSCAN, smallest group you care about	too many tiny groups: raise `min_cluster_size`
Name	a small local LLM on a few samples	names are vague: show more or more varied samples
All in one	BERTopic	you need each step under your own control

SECTION 08 · COMPLETE

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